

```
In [3]: import boto3
import pandas as pd
import io
from sklearn.preprocessing import StandardScaler
```

```
In [5]: bucket = 'cust-seg-ygp'
rfm_key = 'rfm_table.csv' # file you previously saved
rfm_scaled_key = 'rfm_scaled.csv' # output file name to save

s3 = boto3.client('s3')
obj = s3.get_object(Bucket=bucket, Key=rfm_key)
rfm = pd.read_csv(obj['Body'])
print("Loaded RFM shape:", rfm.shape)
display(rfm.head())
```

Loaded RFM shape: (5878, 4)

	Customer ID	Recency	Frequency	Monetary
0	12346	326	12	77556.46
1	12347	2	8	4921.53
2	12348	75	5	2019.40
3	12349	19	4	4428.69
4	12350	310	1	334.40

```
In [7]: # 4) pick columns to scale (DO NOT scale Customer ID)
features = ['Recency', 'Frequency', 'Monetary']
rfm_features = rfm[features]
```

```
In [10]: # check if there are any missing values
print("Any missing values in features?\n", rfm_features.isna().sum())
```

```
Any missing values in features?
Recency      0
Frequency    0
Monetary     0
dtype: int64
```

```
In [9]: # 5) scale using StandardScaler
scaler = StandardScaler()
scaled_values = scaler.fit_transform(rfm_features) # result is numpy array
```

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In [11]: # 6) convert scaled values back to dataframe with readable column names
rfm_scaled = pd.DataFrame(scaled_values, columns=['Recency_scaled', 'Frequency_scaled', 'Monetary_scaled'])
```

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In [12]: # 7) combine scaled columns with Customer ID so each row is still traceable
rfm_out = pd.concat([rfm[['Customer ID']].reset_index(drop=True), rfm_scaled.reset_index(drop=True)], axis=1)
print("Scaled RFM preview:")
display(rfm_out.head())
```

Scaled RFM preview:

	Customer ID	Recency_scaled	Frequency_scaled	Monetary_scaled
0	12346	0.595584	0.438998	5.166378
1	12347	-0.952279	0.131502	0.136127
2	12348	-0.603532	-0.099120	-0.064857
3	12349	-0.871064	-0.175994	0.101996
4	12350	0.519146	-0.406616	-0.181549

```
In [13]: # 8) save scaled rfm to S3
csv_buffer = io.StringIO()
rfm_out.to_csv(csv_buffer, index=False)
s3.put_object(Bucket=bucket, Key=rfm_scaled_key, Body=csv_buffer.getvalue())
print(f"Saved scaled RFM to s3://{bucket}/{rfm_scaled_key}")
```

Saved scaled RFM to s3://cust-seg-ygp/rfm_scaled.csv

```
In [14]: rfm_scaled.describe()
```

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Out[14]:
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	Recency_scaled	Frequency_scaled	Monetary_scaled
count	5.878000e+03	5.878000e+03	5.878000e+03
mean	-4.230860e-17	-1.329699e-17	1.087935e-17
std	1.000085e+00	1.000085e+00	1.000085e+00
min	-9.570564e-01	-4.066161e-01	-2.045035e-01
25%	-8.376226e-01	-4.066161e-01	-1.810036e-01
50%	-5.032078e-01	-2.528681e-01	-1.446135e-01
75%	8.535607e-01	5.462788e-02	-4.900398e-02
max	2.568631e+00	3.011236e+01	4.003090e+01

```
In [ ]:
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